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Request Granularity and Concurrency Adaptation for HTTP Range Downloads under Controlled Server-Side Pacing: A Mechanism-First Evaluation

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ABSTRACT Adaptive HTTP range downloaders choose concurrency (N) and range size (S), but it is rarely clear when adaptation has anything to exploit. We ask whether adapting N and S during a download helps when network or server conditions are not optimal, compared with aria2 and other segmented downloaders, and whether any gain comes from adapting or from how the file is split. On a controlled platform, a fixed configuration of our downloader (16 connections, 4 MiB ranges) completed **62.4 % faster than aria2 1.37** on one characterised workload (30/30 paired runs), a gap that controls attribute largely to request granularity and request-start timing rather than to adaptation. On five replayed cellular traces a counterfactual oracle found no exploitable concurrency headroom (one-sided 95 % lower bound −0.26 % against a predeclared 10 % bar). Under a constructed server whose per-request rate falls with concurrency, a causal controller was 23.4 % faster than the best of three stationary configurations, but a 230-run ablation showed that aria2's lead there came from two engine differences; with both fixed, the best fixed configuration was faster than the controller (1.071 [1.054, 1.085]). Learned policies were slower than aria2 overall, and a retrained DQN beat the best fixed configuration only where it was trained (0.907 [0.893, 0.921]). Against aria2 under changing conditions, adaptive arms were faster only when the server rate rose (0.36–0.88), where a fixed configuration of the same engine was faster still (0.344–0.376), and slower whenever the server degraded (1.096–1.781). All results are scoped to controlled mechanisms, not live networks or production servers.

INDEX TERMS HTTP range requests, segmented downloading, concurrency adaptation, aria2, reinforcement learning, network measurement

1. INTRODUCTION

- Segmented range downloads; fixed vs adaptive N and S. Tools such as aria2, axel and lftp choose N and S once, at the start of a download.
- **Question.** When network or server conditions are not optimal (the server rate changes, the server starts throttling, delay and loss, a link whose capacity swings), does adapting N and S during the download do better than these tools? And is any gain due to adapting, or to how the file is split? To separate the two, every adaptive arm is compared with aria2 and with the best fixed configuration of the same engine.
- Adaptation is useful only when the environment has a state-dependent optimum that a causal controller can observe.
- **What is not new.** Online tuning of parallel streams (FastBioDL, Falcon, direct search), deep RL for transfer parameters, and splitting slow segments (aria2, IDM) are established; we propose no new tuning algorithm.
- Contributions (each with its evidence label):
  1) A controlled, instrumented platform (downloader, server with explicit rate rules, policies, harness) and a claims checker that recomputes every number in this paper from raw artifacts.
  2) A partly mechanism-explained comparison with aria2 on one characterised workload (**Confirmed**, single workload; sensitivity screen in §S4.1), whose direction also held against axel and lftp and across a delay ×

loss grid (screens, §S4.3, §S4.4).
3) A replay-based negative result on recorded cellular traces, with the oracle headroom bounded before any learner was trained (**Confirmed** under the frozen counterfactual model; not a live-network validation).
4) Existence results for adaptive (N,S) selection/control **under an explicitly constructed N-dependent server model** (§S6, §S7): within-download improvement over the best stationary configuration (**Confirmed**, n=10 blocks); cross-workload selection (**Descriptive**; consistency across the family not established).
5) An engine ablation showing that a whole-request timeout and missing straggler splitting, not control, explained aria2's lead under the cap, after which the best fixed configuration beat the adaptive controllers (**Confirmed**, §S7a); and a live-rate control server showing that the large-range advantage depends on the server fixing a response's rate on arrival (§S9c).
6) An inventory of learned-policy experiments (§S9): the early candidates are negative or descriptive; agents retrained for the capped server beat the fixed configuration with their own range size where trained, generalized partially inside a broadened training envelope, and failed a no-harm check on public mirrors (§S8a).
7) A synthesis against aria2 under changing conditions (§S9e).

- Scope: controlled server/network mechanisms; not claims about production servers.

2. RELATED WORK

All entries below were checked against an authoritative record; see `reports/references.md` for the verification log and the list of unresolved candidates, which are deliberately not cited.

Online tuning of transfer parallelism. Balaprakash et al. [1] treat the number of parallel streams as an online optimization problem solved by direct search on production GridFTP/Globus links. Falcon [2], [3] tunes the degree of parallelism with a utility function and online search for high-speed science networks. FastBioDL [4] applies gradient-descent concurrency tuning to HTTP/FTP genomic downloads (arXiv preprint, no peer-reviewed venue found). These works optimize concurrency online against observed throughput; the present paper instead asks, with a controlled server whose mechanism is known, whether an optimum exists for a causal controller to track at all.

Multi-source and range-based downloading. MDTP [5] splits a request into chunk requests across replica servers and sizes chunks to measured per-server throughput, comparing against aria2 [6]. HTTP range requests are specified in RFC 9110 [7]. Our setting is single-source.

Adaptation in adjacent domains. BOLA [8], [9] selects video bitrates by utility maximization without bandwidth prediction and serves as an analogy for decision rules that need no throughput forecast. DeLoad [10] (preprint) sizes download tasks for short-video preloading. Neither is a range-download concurrency study.

Position. The contribution here is a mechanism-first, controlled evaluation with explicit failed gates and a practical baseline comparison, not a new tuning algorithm.

3.0 REPORTING CONVENTIONS

downloads under controlled server-side pacing: a mechanism-first evaluation.*

Evidence labels used throughout. Confirmed = reproduced from raw run files with the stated n; **Descriptive** = point estimate without an inferential claim; **Failed gate** = a predeclared criterion that was not met; **Limitation** = scope or validity caveat. Every number in this draft is recomputed from raw artifacts by `tools/claims_check.py`.

Note on "pre-registered". Protocols in this project were written to logged documents with recorded SHA-256 hashes before the corresponding runs. They were not deposited with an independent timestamping service, the repository history begins with a single import commit, and one manifest was amended twice after design output had been seen (W1–W6, §S6.1; the first version was overwritten). Wording such as "pre-registered" in older internal documents therefore means predeclared in a logged protocol, not independently verifiable pre-registration. See `docs/PREREGISTRATION_NOTE.md`.

3. SYSTEM AND EXPERIMENTAL PLATFORM

- Go downloader: RangeTracker (half-open), fetcher (inclusive), assembler, metrics, worker pool (dynamic N, in-flight ranges immutable), concurrency and chunk-size controllers, policy interface, serialized policy runner.
- Controlled server: deterministic logical object; per-response pacing sampled at **request start and held for that response**; wall-clock rate schedules; fault injection; and a soft concurrency cap `rate_eff = min(base, C/(k*(1+gamma*(k-1))))` evaluated at request start with k = concurrent requests including the arriving one (C = 8 MiB/s and gamma = 0.05 were chosen in an exploratory screen of 27 schedule–cap pairs, of which one met the selection rule). This request-start semantics is a property of the server model and drives the aria2 results below. An opt-in live-rate mode instead re-evaluates each response's rate during the response from the current k; it serves as a control in §S9c.
- Harness: experiment matrix, aria2/curl/wget runners, SHA-256 content verification of every output, tc/netem [11], and Mahimahi mm-link trace replay [12].

4. PHASE-LOCKED PACING AND THE ARIA2 COMPARISON

- Workload D2: 512 MiB, per-response pacing 640 → 160 → 2500 KiB/s at 2 s / 4 s, netem `delay 40ms loss 0.4%` on loopback, which applies in both directions (measured RTT 80 ms; earlier drafts said "40 ms RTT"); both systems receive the same object.

- Frozen configurations (selected on separate discovery runs): ours (16 conn, 4 MiB); aria2 1.37 `–max-connection-per-server=16 –split=16 –min-split-size=1M`. The full aria2 split frontier (1/2/4/8/16) and a 40-configuration grid for ours were run in discovery.
- **Confirmed (30 fresh pairs):** ours 20.12 s vs aria2 53.56 s mean; geometric ratio 0.3756, 95 % CI [0.3726, 0.3787], 30/30 wins; all runs content-verified.
- **Mechanism (per-request server logs, per run):** aria2 issues 16 requests (one unranged probe plus 15 × 32 MiB ranges), all starting before 2 s and held at 640 KiB/s; ours issues 128 × 4 MiB ranges, of which 16 start before 2 s and 112 start after 4 s in the recovered phase (3840 ranged requests, 480 early / 3360 late, over 30 runs). aria2 cannot be configured to finer initial pieces: with `–x16`, `–split` of 16/64/256 gave the same range-size distribution in a local probe.
- **Controls:** granularity-matched control ratio 0.862 [0.834, 0.890] (10 pairs). A constant 2.5 MB/s schedule removes most of the 4 MiB vs 32 MiB difference within our own downloader (2.3× → 1.1×) but ours is still faster than aria2 (15.3 s vs 24.8 s); see §S4.1. The phase-locked mechanism therefore explains the size of the D2 gap, not the whole of it.
- **This is a fixed-configuration result, not an adaptive one:** neither system adapts during the transfer in this comparison.
- **Limitation:** one workload family on a synthetic server; the win is a property of this pacing mechanism, not a general statement about aria2.

4.1 PACING-SCHEDULE SENSITIVITY (SCREENING)

Frozen protocol: `testdata/d2_pacing_sweep/manifest/sweep_manifest.json` (committed before any run; order seed 20261011; exclusion rules fixed). Same workload, systems, and netem as §S4 (512 MiB; ours 16 conn × 4 MiB; aria2 frozen flags); only the server pacing schedule varies. **Screening only: five pairs per schedule, no inferential claim.** 60/60 runs valid, 0 exclusions, all content-verified.

Both predeclared expectations held: S0 reproduces the historical D2 ratio (0.378 vs 0.3756) and S5, where both systems run 16 connections at one slow rate, is at parity. Beyond that the screen is descriptive: the advantage is largest when the early slow phase is short relative to the transfer (S0, S1), shrinks when the slow phase lasts longer (S2, 0.806), and persists under constant fast pacing (S4, 0.568), where no phase-locking exists. The sweep did not test why S4 favours the smaller ranges (an obvious candidate is the finishing tail of 32 MiB pieces, unmeasured here), so the mechanism for S4 is a hypothesis, not a finding. S1 has one pair at 0.751 (a slow ours run); it is retained under the predeclared exclusion rules.

4.2 STOCK-SERVER CHECK (NGINX `LIMIT_RATE` / `LIMIT_RATE_AFTER`)

Purpose and scope: a separate screen of whether the direction of the aria2 comparison survives on a stock server whose rate control we did not write. It is **not** a shared or concurrency-dependent cap: NGINX 1.24.0 `limit_rate` limits each response independently and `limit_rate_after` leaves the first bytes of each response unthrottled; there is no concurrency-dependent behaviour and no connection limit. Frozen protocol: `testdata/nginx_limit_rate/manifest/nginx_manifest.json` (committed before any run). Same 512 MiB object (deterministic AES-CTR stream, SHA-256 verified), netem, systems and aria2 flags as §S4; 5 pairs per configuration; screening only. 20/20 runs valid.

The predeclared expectation for N1 was parity (both systems use 16 connections at one per-response rate, as in sweep schedule S5, which gave 0.998). **That expectation was not met**: ours was faster in all five pairs. The check does not identify why; the difference between S5 and N1 could come from the server implementation (NGINX/sendfile pacing granularity), aria2's request handling against NGINX, or the two systems' connection behaviour, and none was tested. The result supports only the statement that the direction of the comparison (ours faster) was also observed on a stock server under per-response rate control in this setup.

4.3 OTHER SEGMENTED DOWNLOADERS (AXEL, LFTP)

A frozen screen (`testdata/baselines_ext/protocol.md`) added axel and lftp `pget` next to aria2, all at 16 connections, on the same server models and netem. HEAD requests were answered at once, so no tool paid for a paced probe. 5 blocks per scenario, 80 runs, all valid. wget2 was planned and then removed (amendment A2): it fetches one file over a single connection however it is configured. We confirmed this in its source; upstream issue

#195 describes the same behaviour.

- **Reading.** axel and lftp split the file into 16 equal parts at the start, as aria2 does. They gain nothing from a pacing schedule that rises after the first second, so the §S4 gap is not specific to aria2: it applies to tools that split the file once, up front.
- **Under the cap, all four were level** at about 53 s. With the cap from the start, lftp took 114 s because it never re-splits a slow part (our engine without splitting behaved the same way, §S7a), and axel was 2.5 % faster than ours.
- **Screen only:** 5 blocks on constructed server models.

4.4 DELAY AND LOSS SWEEP (SCREEN)

A frozen sweep (`testdata/netem_sweep/protocol.md`) repeated two comparisons over a grid of one-way delays (10, 40, 100 ms) and loss rates (0, 0.4, 2 %). It used 3 blocks per cell, 108 runs, all valid.

schedule (KiB/s; switch times s)	ours (s)	aria2 (s)	geo ratio ours/aria2	[min, max]	ours faster
S0 D2 reference `640,160@2,2500@4`	20.1	53.3	0.378	[0.372, 0.389]	5/5
S1 no dip `640,2500@2`	20.1	48.9	0.425	[0.353, 0.751]	5/5
S2 long dip `640,160@2,2500@8`	44.1	54.7	0.806	[0.783, 0.827]	5/5
S3 partial recovery `640,160@2,1280@4`	31.6	54.3	0.583	[0.559, 0.608]	5/5
S4 constant fast `2500`	15.4	27.1	0.568	[0.523, 0.618]	5/5
S5 constant slow `640`	53.8	53.9	0.998	[0.987, 1.008]	3/5

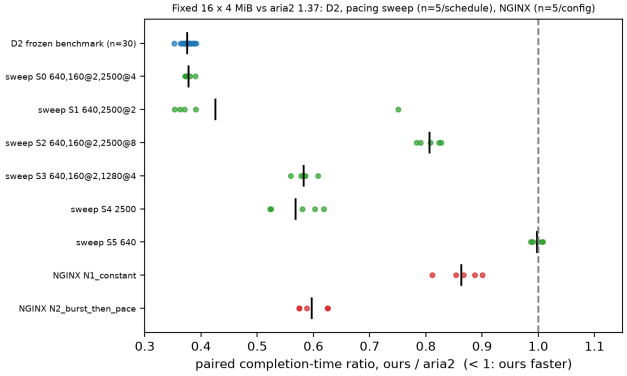


FIGURE 1. Paired completion-time ratios ours/aria2 for the frozen D2 benchmark, the six sweep schedules and the two NGINX configurations. Dots are pairs; bars are per-cell geometric means; the dashed line is parity. Sweep and NGINX cells have n=5 pairs and are screening evidence only.

configuration	ours (s)	aria2 (s)	geo ratio ours/aria2	[min, max]	ours faster
N1 `limit_rate 640k` (constant per response)	50.3	58.3	0.863	[0.811, 0.901]	5/5
N2 `limit_rate_after 1m; limit_rate 640k`	33.7	56.4	0.597	[0.574, 0.625]	5/5

Scenario	Ours (setting)	ours ÷ axel	ours ÷ lftp	ours ÷ aria2
HEALTHY (t1, no cap)	16 × 1 MiB	0.345 [0.344, 0.347]	0.345 [0.344, 0.346]	0.357 [0.344, 0.371]
D2	16 × 4 MiB	0.375 [0.373, 0.378]	0.374 [0.369, 0.377]	0.375 [0.373, 0.377]
CAP8	16 × 32 MiB	1.003 [0.965, 1.026]	1.001 [0.963, 1.024]	0.996 [0.960, 1.017]
CAP0	16 × 32 MiB	1.025 [1.021, 1.029]	0.597 [0.594, 0.599]	0.995 [0.992, 0.998]

- **D2 (16 × 4 MiB against aria2):** ours was faster in every block of all 9 cells, with cell geometric means from 0.362 to 0.457. The direction holds across the grid, by the predeclared rule.
- **CAP8 (16 × 32 MiB against aria2):** the direction does not hold.
  - 5 cells were mixed, and 2 had aria2 faster in every block: 10 ms with 2 % loss (1.073) and 40 ms with 0.4 % loss (1.035).
  - At 100 ms, ours was faster with 0 and 0.4 % loss (0.769 and 0.955) and mixed with 2 % loss.
  - So under the cap the two are near parity, and the sign depends on the network.
- **Screen only:** 3 blocks per cell on one machine. No claim about real networks.

5. WHEN DOES ADAPTATION HELP? MECHANISM SCREENS

Predeclared screens, each closed before policy work: static rate+RTT (chunk size moves, no causal switching gain; analytical), live queueing (work-conserving; no joint opportunity), correlated loss (no joint opportunity; N monotone. An earlier "S fragile" reading was an artefact of a modelled 30 s request deadline: re-simulated without it, S* = 16 MiB in all 9 variants, with no failures), failure-based N-dependent throttling (abort cliff), and adaptive-N degradation (interior optimum analytically; live mismatch). Existing adaptive heuristics on D2 adapt but lose to the fixed optimum (joint 2.32× slower). **Conclusion:** within-download adaptation needs a mechanism whose optimum moves in N and S and is causally observable; none was found live in the screened mechanisms other than the constructed soft cap of §S6–§S7. The same screens searched a space of server behaviours

(72 cap candidates in §S6.1 alone); results in §S6–§S7 are existence demonstrations in a selected environment, not prevalence estimates.

6. CROSS-WORKLOAD SELECTION UNDER THE CONSTRUCTED CAP

- Family: healthy server vs concurrency-capped server (C=8 MiB/s, gamma=0.05); stationary optima (16, 1 MiB) vs (4, 4 MiB); cross penalties +35.8 % / +252.9 %.
- Live mechanism gate (**Failed gate**; five of six predeclared checks passed): N* moved 16→4 and S* moved 1→4 MiB (N-margin 5.75 %, just above its 5 % bar), and cross-state penalties were material (+31.8 % using the healthy optimum under the cap; +254.4 % using the capped optimum when healthy). The local S-margin was 3.73 % in the gate grid (3.74 %, 95 % bootstrap [3.09 %, 4.45 %], after four extra replicates) against a predeclared 5 % criterion; the criterion was not changed. Three high-N/large-S capped cells aborted at the 30 s request timeout and are preserved as invalid (a 20/30/60 s timeout screen left the optimum at (4, 4 MiB) and changed only reachability).
- Selector: probe at (16, 1 MiB), classify after $\geq$ 20 samples, apply the predeclared (N,S), no further switching. Live, 10 blocks per workload: healthy 19.1 s vs fixed (16,1M) 19.0 s (geo 1.006 [0.931, 1.082], 6/10) and vs aria2 53.9 s (geo 0.353, 10/10); capped 84.2 s vs fixed (16,1M) 115.3 s (geo 0.730, 10/10) and vs aria2 68.3 s (geo 1.232, 0/10: aria2 faster).
- Baselines, stated explicitly: vs the best single fixed configuration across the two workloads (8, 4 MiB; 36.3 s / 91.7 s) the selector gains +47.5 % / +8.2 % (**Descriptive**; the two workloads were designed to have different optima); vs the per-workload best fixed it is −3.5 % healthy and +2.6 % capped.

6.1 SIX-WORKLOAD FAMILY

Frozen manifest `testdata/n_dependent_server_joint_ns/manifest/w1_w6_manifest.json` (predeclared in a logged protocol, but not independently timestamped): six predeclared workloads (W1 healthy; W2–W6 cap/schedule/transition variants) chosen by a predeclared qualification and selection rule from 72 analytic candidates (13 qualified). Two post-hoc amendments were made after design output and before any live W1–W6 run (post-drop window 20 samples → fixed 25 s; relaxed transition coverage); the first manifest was overwritten and is not preserved. W2–W4 share one selector target by construction. Live fixed-grid validation: 5/6 model optima matched; **W5 mismatched** (live argmin (16,16M) 85.4 s; the model's target (4,4M) measured 95.9 s live against a model prediction of 94.8 s) and the frozen selector was not retuned. 10 fresh paired blocks per workload, 240 runs, 0 invalid.

Primary comparison is the workload-level interval, because the unit of generalisation is the workload, not the block. It includes 1 for all three comparisons. The 0.770 point estimate is therefore **Descriptive**: a consistent improvement over a single family-wide configuration across the family is **not established**. It is driven by W1 (healthy workload vs a degraded-state fixed configuration; ratio 0.286); over W2–W6 the geometric mean is 0.94 and the selector loses on W5 (1.058) and W6 (1.090).

aria2 is faster on every degraded workload (its burst-started 32 MiB pieces start before the cap rate drops, §S7). The result is workload-level selection under the constructed cap, not within-download adaptation.

7. WITHIN-DOWNLOAD ADAPTATION UNDER SERVER DEGRADATION

- State schedule: no cap on [0, 8 s), cap on [8 s, EOF). Controller: running max of windowed median per-request throughput; on a drop below 0.5×, switch once to a predeclared target ((4,4M) joint; (4,1M) N-only; (16,4M) S-only).
- **Confirmed (10 paired blocks, 0 invalid):** causal joint 58.7 s vs best of the three stationary configurations run under this schedule, (16,1M) 76.7 s: geo 0.766 [0.762, 0.769], 10/10 (+23.4 %). The comparison set is {(16,1M), (4,1M), (4,4M)}; no fixed grid was run under this schedule, so "best" means best of those three.
- Ablation: N-only 62.0 s (geo 0.808 vs best fixed, 10/10); joint vs N-only geo 0.947 [0.943, 0.952], 10/10; S-only 82.1 s, worse than not switching (geo 1.070, 0/10). Under this cap the N move carries the gain.
- **aria2 is faster:** 53.6 s vs causal joint 58.7 s (geo 1.096, 0/10). This result supports improvement over the best stationary configuration under the selected controlled transition, **not** superiority over aria2. Mechanism (per-request server logs): aria2's first nine pieces start at k $\leq$ 9 where the cap rate exceeds the base rate and hold 640 KiB/s for ~52 s; the server fixes a request's rate at its start.
- Block-level CIs here are very narrow because the emulated server is nearly deterministic; they quantify emulator noise, not population variability.
- **Superseded explanation of the aria2 gap.** §S7a shows that the gap came mainly from two engine details of our downloader, not from aria2's piece timing alone, and that against the best fixed configuration on the corrected engine the causal controller is slower.

7A. ENGINE ABLATION: WHY ARIA2 LED UNDER THE CAP

An audit found two engine differences between our downloader and aria2: (a) every request had a 30 s whole-request deadline, which aborts a large range that is slow but progressing, and (b) in-flight ranges were never split, while aria2 lets an idle connection take half of the slowest segment's remaining bytes.

comparison (selector vs ...)	pooled geo ratio	block-level 95 % CI	workload-level (six-cluster) interval	blocks won
per-workload best fixed	0.999	[0.969, 1.029]	[0.873, 1.143]	24/60
family-best single fixed (4, 4 MiB)	0.770	[0.684, 0.859]	[0.455, 1.304]	40/60
aria2	1.660	[1.360, 1.990]	[0.707, 3.901]	10/60

workload	vs per-workload best fixed	vs family-best fixed
W1	1.013	0.286
W2	0.934	0.869
W3	1.019	0.949
W4	0.808 (selector runs the fast (16,1M) config until the cap appears: probe-phase timing, not better selection)	0.765
W5	1.172 (frozen-target mismatch)	1.058
W6	1.088	1.090

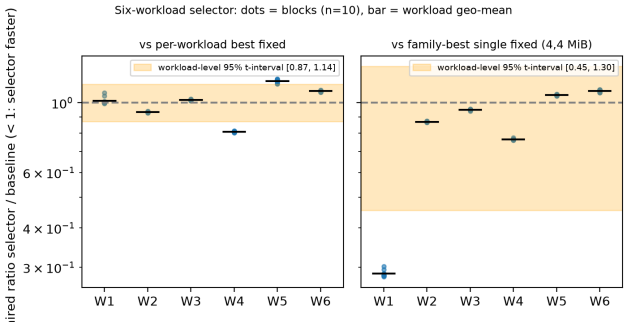


FIGURE 2. Selector paired ratios per workload against the per-workload best fixed configuration (left) and the family-best single fixed configuration (right). Dots are blocks; the shaded band is the workload-level 95 % t-interval (six clusters), which includes 1 in both panels.

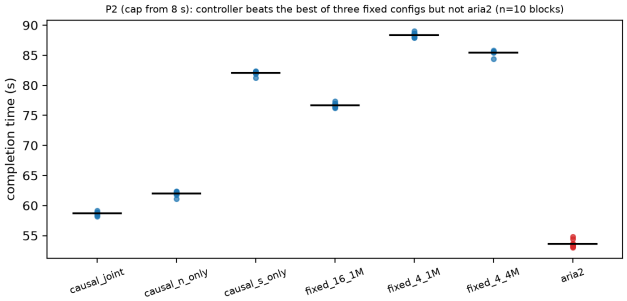


FIGURE 3. P2 completion times (10 paired blocks). The causal joint controller is faster than the best of the three fixed configurations run under this schedule but slower than aria2; the S-only ablation is worse than not switching.

Both were made opt-in options (`NoRequestTimeout`, `SplitStragglers`, default off; a split needs at least 2 MiB unreceived) and tested in a predeclared ablation (`testdata/split_ablation/protocol.md`): four scenarios (cap from the start, CAP0; cap from 8 s, CAP8; no cap; the D2 pacing workload) × 10 shuffled blocks, 230 runs, all valid, with zero failed attempts, retries or wasted bytes on the client side. (The server logs do show aborted requests. These are the unreceived tails of ranges that were split, cut off when the client closes them; the client discards no bytes it received.)

- **Splitting was the main cause (Confirmed, 10 blocks).** Under CAP0 the same 16 × 32 MiB configuration took 40 % less time with splitting (0.599 [0.598, 0.601]) and then matched aria2: 68.2 s against 68.5 s, 0.995 [0.992, 0.999], detectable but practically negligible.
- **Large ranges need the deadline removed (Confirmed).** Under CAP8, 16 × 32 MiB without the deadline took 0.711 [0.703, 0.719] of the time of the earlier best fixed configuration (16 × 1 MiB), and was within 1.6 % of aria2 (54.4 s against 53.5 s, 1.016 [1.006, 1.029]).
- **No harm elsewhere.** With no cap and on D2 the corrected and original engines were not distinguishable (0.981

[0.931, 1.011]; 0.995 [0.987, 1.003]), and ours remained much faster than aria2 there (0.344 and 0.370).

- **Adaptation against the corrected baseline.** The causal controller of §S7 replicated against its original comparison set (0.767 [0.764, 0.769]), but against the best fixed configuration on the corrected engine (16 × 32 MiB) it was slower (1.071 [1.054, 1.085]); under CAP0 the workload selector of §S6 was slower than 16 × 32 MiB (1.229 [1.223, 1.234]).
- **Mechanism.** The server fixes a response's rate when the request arrives. Under CAP8, 16 × 32 MiB starts all 16 ranges before the cap, so every range keeps the pre-cap rate; splitting then hands the tail to idle connections. This is a property of our server model; a server that re-evaluates rates during a response could change it.

7B. ONLINE CONCURRENCY TUNING (PRIOR-WORK RE-IMPLEMENTATION)

FastBioDL-style online tuning [4] was re-implemented as a project policy (not their code): aggregate throughput T over 2 s windows, utility U = T/K^N, keep the step direction while U improves, reverse and halve on a drop above 5 %, clamp at N in 1–16, S untouched. Predeclared protocol (`testdata/online_tuner/protocol.md`): CAP8 and no cap, 8 shuffled blocks, 64 runs, all valid, corrected engine.

- **Slower than the better of the two fixed 1 MiB configurations in both scenarios (Confirmed).** CAP8: 1.212 [1.180, 1.248] against 16 × 1 MiB; no cap: 1.220 [1.203, 1.232]. The tuner keeps the range size at 1 MiB, so these are same-range-size comparisons; under CAP8 the corrected engine's best fixed configuration (16 × 32 MiB, §S7a) was faster still.
- **Adaptation speed (Descriptive).** It cut N 2.1 s after the cap (the causal controller 1.5 s), but needed 8.1 s to climb from N = 4 to N = 16 without a cap, and that ramp is where most of the loss occurs.

8. RECORDED CELLULAR TRACES REPLAYED IN A CONTROLLED SETUP

Five recorded driving traces (Verizon EVDO and LTE, AT&T LTE, T-Mobile LTE and UMTS), recorded with the Saturator tool of Winstein et al. [13] and distributed with Mahimahi [12], were replayed through mm-link on a controlled host; this is trace replay, not measurement on live mobile networks. A per-trace counterfactual oracle with transition costs estimates the completion-time headroom of switching concurrency over the best fixed C. Headroom per trace: 1.64 %, 0.019 %, 0.016 %, 0.061 %, 0.294 % (mean 0.406 %). The predeclared gate required a one-sided 95 % lower bound ≥ 10 %; the bound was −0.26 % (**Failed gate**; n = 5, df = 4). Oracles are not policies; this bounds the benefit available to any concurrency-switching policy under the frozen model.

What the traces vary. Each trace lists the times at which the recorded link could deliver one 1500-byte packet. In 1 s bins, downlink capacity swings by a 90th/10th percentile ratio of 6.5 to 25.3 across the five traces (Verizon EVDO 24.2, Verizon LTE 14.1, AT&T LTE 6.6, T-Mobile LTE 6.5, T-Mobile UMTS 25.3; `tools/trace_swing.py`). The replay ran `mm-link` with the trace alone, without `mm-delay` or `mm-loss`, so it varied capacity only: no added propagation delay, no random loss, and no server behaviour. Loss and delay were studied separately with netem (§S4.4, §S5). On one shared bottleneck whose capacity varies, the best N stays roughly "enough to fill the pipe", because every setting slows down together. A capacity trace cannot express a per-connection cost that changes over time, which is the mechanism §S6–§S7 construct.

8A. LIVE-INTERNET PILOT (DESCRIPTIVE)

To check whether the trace-replay conclusion survives contact with real paths, we ran a small descriptive pilot (`testdata/live_pilot_l1/`; protocol committed before the first slot, with amendments logged). One client on the author's residential broadband in India downloaded a kernel.org release tarball from four public mirrors: the kernel.org CDN, the kernel.org origin mirror, and university mirrors in Japan and China. Each download used one of eight fixed (N, S) cells (N in {1, 4, 8, 16}, S in {1, 4} MiB), and aria2 and a single-stream curl reference ran alongside. There were two reps per cell per server in four time-of-day slots on the primary access network (Sunday midday, Sunday evening, Monday evening, Tuesday morning), plus a re-run of the midday slot for the Tsinghua mirror (amendment A2). Every file was checksum-verified and invalid runs were kept. A condition is one (server, slot) pair.

- **Validity:** 328 of 350 runs valid. The invalid runs are the 16 downloader runs that the Tsinghua mirror refused with HTTP 403 in the first slot (see below), and 6 runs during an internet drop at the start of the first Tuesday attempt, which was then re-run in full (amendment A4). The fastest valid run reached 2.43 MB/s, which is the access link's ceiling.
- **Best fixed cell per condition:** no single cell was top everywhere, but the top cell always had N of 8 or more, except for 4 × 1 MiB on the CDN in two slots and 4 × 4 MiB on the Tsinghua mirror in one. The top cell changed across slots on 4 of 4 servers. How many connections were "enough" depended on the server. On the origin mirror, the better of the two N = 4 cells was 1.37–1.71× slower than that condition's best in every slot, and N = 1 was slower still.
- **Selection headroom:** one global cell, 8 × 1 MiB, was on average 3.7 % slower than each condition's clairvoyant best (95 % bootstrap over conditions [1.0, 4.8]; 16 conditions). It was within 5 % of the best in 11 of 16 conditions and at most 15.3 % slower in any one, with two reps per cell. No condition was access-link-bound by the protocol's definition, because N = 1 was always more than 5 % slower than the best, so the headroom excluding

such conditions is identical.

- **Within-run variability:** median coefficient of variation of 1 s throughput bins 0.11–0.29 across servers.
- **Context:** aria2's time ratio over the condition's best fixed cell was 1.00–1.26.

In this pilot, one fixed configuration (8 × 1 MiB) was near-best across all observed conditions: the bootstrap upper bound of 4.8 % is below the predeclared 5 % line. This supports, but does not prove, the thesis that real-condition variability did not move the best setting. The bound is close to the line, and individual conditions deviated by up to 15.3 %, so this is a screening result. A side observation: the Tsinghua mirror rejected the Go standard library's default User-Agent with HTTP 403 for every range request, while accepting curl, aria2 and an identifying User-Agent. The downloader's runs against it therefore used an explicit User-Agent (amendment A2).

One morning slot ran by accident on a second, unidentified access network (amendment A3). It is reported separately and is never pooled with the primary conditions. On that network 68 of 80 runs were valid. The failures were DNS and retry exhaustion, clustered in about half an hour of unstable connectivity. Peak throughput was 12.3 MB/s, and within-run variability was far higher (median coefficient of variation 1.16–1.77). The same global cell, 8 × 1 MiB, had 7.2 % headroom (bootstrap [1.2, 12.2]; 4 conditions; within 5 % of the best in 2 of 4). This single slot does not meet the predeclared small-headroom criterion. With four conditions from one partially unstable slot it is inconclusive, but it is the one place in the pilot where the best setting may have moved by more than 5 %.

The pilot has one client and one primary access network. Its slots span about two days, and CDN and mirror behaviour may change over time. It is screening evidence that a fixed configuration with enough connections was near-best on the observed paths. It does not validate the findings across networks.

Adaptive arms on the same mirrors (descriptive no-harm check, `testdata/live_mirror_check/protocol.md`). After the loopback studies, the v2 agent of §S9d and the C=8-tuned drop rule ran on three of these mirrors (kcdn, kedge, riken). Each was compared with fixed 16 × 1 MiB and 8 × 1 MiB: the same corrected engine, 5 blocks, 64 MB.

- **The first slot was lost to the client's own link.** From 22:10 UTC every attempt to all three servers failed with no response, and only 20 of 60 runs were valid. It is reported in full and not used (amendment A2).
- **The second slot ran after a predeclared link check, and 60 of 60 runs were valid.** v2 against the better fixed arm:
  - kcdn: 1.401 [1.102, 1.773], slower;
  - kedge: 1.654 [0.873, 2.849];
  - riken: 1.215 [0.719, 1.899]. The drop rule against the better fixed arm:
    - kcdn: 1.125;
    - kedge: 1.063;
    - riken: 1.332. Only the kcdn v2 interval excludes 1. Under the frozen rule, "no harm" needs a ratio of at most 1.05 that is not slower, and both arms showed no harm on 0 of 3 mirrors.
- **What happened.** The mirrors have no cap, but both arms cut their connections: v2 ended at a median N of 6–7 on kcdn and kedge, and the drop rule switched to N = 4 in 4 of 5 runs on kedge and on riken. Our reading, not tested against the throughput traces, is that noisy per-request throughput looked like a cap.
- **Reading.** The agents adapt to signals that, on these servers, do not mean what they meant on the server model they were built for. Adaptation carried a cost here, with little selection headroom to pay for it (3.7 % in the pilot). This matches §S8, where variability did not mean opportunity. It is one client, one evening and 5 blocks of a noisy link, so it is descriptive only.

9. LEARNED-POLICY EXPERIMENTS (INVENTORY; POSITIVE RESULTS ONLY IN 9B AND 9D, BOTH SCOPED)

The learners are a LinUCB contextual bandit [14] and Double DQN [15] agents in the DQN family [16], all acting through the same serialized policy interface as the deterministic controllers.

Naming. "DQN v1–v6" in the table below are the early candidates. The agents of §S9b and §S9d are a separate pair retrained for the capped server, called the cap-trained v1 and cap-trained v2: the cap-trained v2 of §S9d is not the early DQN v2 of the table.

Status column: **valid** = the stated comparison is internally consistent and reproducible from its artifacts; **stale** = produced before a later correction of its simulator/environment; **non-comparable** = different workload, file size or aria2 configuration from the other rows, so ratios must not be pooled.

No learned policy outperformed the practical baseline in a supported comparison. Two features of these head-to-heads worked against the learned policies regardless of what they learned:

- **Short downloads.** The files were 4–16 MiB, so most runs lasted only a few seconds.
- **Ramped start.** v6 started at one connection and ramped up, while aria2 opened all of its connections at once.

The startup cost is therefore a large share of each run. The long-duration diagnostic shows the pooled gap shrinking with file size, but it does not remove the gap per trajectory. These comparisons test the frozen candidates as built; they do not show that a learned controller cannot match aria2. The single sub-1 pooled value (v6 at 512 MiB) has an interval that includes 1 and is driven by one trajectory. Candidate artifacts are frozen with SHA-256 hashes; none was tuned after its evaluation. Intervals in this table are

experiment	design	outcome	status
LinUCB, five- and six-mode 4 MiB benchmarks	300 runs, 5 scenarios, fixed/rule-based/LinUCB modes	descriptive; locks onto a single arm after warm-up; periodic exploration changes coverage but not the dominant preference; no ranking made	valid (descriptive), non-comparable to aria2 rows
DQN v1 (frozen candidate)	simulator train/validation/test + 25-run live transfer, no aria2 comparison	superseded; trained on the v1 environment later corrected (§S22.24)	stale
DQN v2 vs aria2 baseline v1	240 pairs, U1–U8 profiles, 4 MiB, aria2 `-x4 -s4`	75.7 % slower (geo 1.757 [1.672, 1.847]); faster in 31/240 pairs	valid; non-comparable to the 16 MiB rows
DQN v3	simulator + live transfer vs v2	fixed fragmentation, but simulator rewarded chunk sizes the server does not reward	valid (negative/mixed)
DQN v4 (concurrency-only, fixed 1 MiB)	simulator + 25-run live transfer, no aria2 comparison	live behaviour matched simulator; 4 MiB file gave only a four-request control horizon	valid; no head-to-head
DQN v5 vs aria2-F	240 pairs, 16 MiB, U1–U8	geo 2.322 [2.230, 2.417]; faster in 1/240 pairs; outperformance criteria not met	valid as a head-to-head; **candidate artifact stale** with respect to a later simulator window-accounting fix (not retrained)
DQN v6 (concurrency 1–16) vs aria2-F	150 pairs, trajectories A2–E2, 16 MiB	pooled geo 1.456 [1.305, 1.624]; faster in 37/150; one of five trajectories not slower (E2, 0.948 [0.897, 1.002]); D2 5.47× slower	valid; non-comparable to v5 rows (different bounds/trajectories)
v6 long-duration diagnostic	16/128/512 MiB, 5 pairs/cell, A2/D2/E2	pooled 1.805 → 1.062 → 0.909 (512 MiB CI [0.750, 1.102] includes 1), driven by D2 (6.00 → 1.16 → 0.568) while A2 and E2 worsen with size	valid but exploratory, small n

95 % t-intervals on the log ratio recomputed by `tools/claims_check.py` with the critical value for the actual degrees of freedom; the frozen analysis script (`python/analyze_final_evaluation.py`) used 2.045 for every n (and the frozen D2 analysis used the normal value 1.96, giving [0.3728, 0.3785] instead of the t-interval reported here), which is correct for n=30 and slightly conservative for n=150 and n=240 (and too narrow for n=15), so its stored intervals differ in the third decimal. The frozen artifacts were not modified.

9A. FROZEN DQN V6 ON THE CAPPED SERVER

To remove the short-file and ramped-start handicaps above, the frozen v6 agent was run on the capped server with a 512 MiB file on the corrected engine, starting at N = 1 (as trained) or N = 16 (as aria2 opens), in a predeclared 5-block screen (`testdata/rl_capped/protocol.md`; 50 runs, all valid). It was not retrained.

- With the cap from 8 s, starting at 16 it was slower than the best fixed configuration (16 × 32 MiB): 1.356 [1.252, 1.472]; starting at 1, 1.706 [1.612, 1.806]. Without a cap it was slower than 16 × 1 MiB (1.031 [1.010, 1.063]; start 1: 1.194 [1.176, 1.219]).
- Against 16 × 1 MiB, the fixed configuration with its own range size, it was not distinguishable under the cap (0.931 [0.879, 1.016]).
- **Descriptive (screen).** The frozen agent did not beat a well-chosen fixed configuration even with a fair start, a long file and a scenario built so that the best setting changes.

9B. DQN RETRAINED FOR THE CAPPED SERVER

A predeclared study (`testdata/rl_cap_train/protocol.md`) retrained the same learner, with the same action space and state layout as v6, on a new simulator of the capped server (`python/rl/sim_env_cap.py`).

- **What the v6 simulator got wrong.** It had no cap fixed at request arrival. Its aggregate-throughput observable also did not match the live runner's: live, it is the bytes of the last `-window` attempts over their span. With window 4 that value barely changes above N = 4, so the new agent uses window 16 in both training and live runs.
- **Calibration.** Per-request server excess and client overhead are drawn from the rl_capped request logs. Completion times were not used.
- **Blind live check (Confirmed for the simulator, 8 runs).** Four fixed settings were predicted before they were run, and all four came within 1.4 % (the largest error was −1.35 %).
- **In distribution (Confirmed, 10 blocks, 120 runs, all valid).**
  - With the cap from 8 s, the agent starting at N = 16 took 0.907 [0.893, 0.921] of the time of fixed 16 × 1 MiB, the best fixed setting with its own range size. The simulator had predicted 0.912.
  - It remained slower than 16 × 32 MiB (1.298 [1.274, 1.321]) and aria2 (1.310 [1.287, 1.331]), as the protocol stated in advance: those large ranges all start before the

cap and keep the pre-cap rate, which a 1 MiB agent cannot do.
  - Against frozen v6 it was not distinguishable (0.949 [0.889, 1.012]).
  - Without a cap it was slower than 16 × 1 MiB (1.108 [1.093, 1.121]).
  - Starting at N = 1 it was slower in both scenarios.
- **Held out (Confirmed, 5 blocks per scenario, 65 runs, all valid).** Three live scenarios each lie outside the training ranges on one axis.
  - The agent was faster than the best fixed 1 MiB setting in none of them.
  - Cap C = 16 MiB/s: 1.177 [1.074, 1.245], slower.
  - 80 ms each way: 1.016 [0.998, 1.043], not distinguishable. The simulator had predicted 0.930; its jitter was calibrated at 40 ms.
  - The d2 schedule, against 4 × 1 MiB: 1.015 [0.998, 1.039], not distinguishable.
- **Reading.** The retrained agent beat the best fixed setting with the same range size on the server model it was trained for. It did not generalize to held-out conditions of the same family, and it did not beat the best fixed setting overall. This is one trained candidate; the result says nothing about real servers.

9C. JOINT CONCURRENCY AND RANGE-SIZE AGENT

A predeclared study (`testdata/rl_joint/protocol.md`) let the same learner also set the range size: 5 actions (none, ±1 connection, ±4 MiB), from 1 to 33 MiB. The aim was to test whether a learned agent finds the large-range behaviour that wins under the cap.

- **Headroom existed in the simulator.** A hand-written rule scored 0.352 of the best fixed time under CAP8: 1 MiB ranges until the rate rises, then 33 MiB ranges.
  - That headroom comes from the server's arrival-time rate rule. Large ranges that start before the cap keep the faster rate for their whole length.
  - The protocol therefore added a control: a server mode that re-checks the rate during each response.
- **Simulator check (Confirmed, 8 blind runs).** Four new fixed settings came within +1.5 % to +7.0 % of their predictions. The simulator is optimistic for long ranges.
- **Training.**
  - The first run was undertrained: its ε schedule ended near 0.75 (amendment A1). The retrain ran 747,503–823,821 steps per seed.
  - Even so, the selected agent scored 1.513 of the best fixed time under CAP8 in the simulator. It did change the range size, but too late.
- **Live, arrival-rule server (Confirmed, 10 blocks, 150 runs, all valid).** The agent was slower than the best fixed setting in every scenario:
  - CAP8: 1.471 [1.405, 1.546] against 16 × 32 MiB (the primary comparison);
  - HEALTHY: 1.081 [1.037, 1.116] against 16 × 1 MiB;
  - CAP0: 1.359 [1.321, 1.399] against 16 × 32 MiB;
  - D2: 1.372 [1.350, 1.397] against 16 × 5 MiB.
  - Against aria2 it was slower under CAP8 (1.422 [1.367, 1.477]) and faster without a cap (0.388 [0.375, 0.404]).
  - Live times were within about −1 % to +5 % of the frozen simulator predictions. At 8 s the agent was still on 1 MiB ranges in every scenario.
- **Live-rate control (Confirmed, 5 blocks, 40 runs, all valid).**
  - Without the arrival rule, fixed 16 × 32 MiB under the cap slowed from 50.9 s to 67.4 s.
  - The agent was not distinguishable from it (1.029 [0.989, 1.081]).
  - Without a cap the agent was slower (1.128 [1.020, 1.259]).
- **Reading, as frozen:** a joint agent trained on a checked simulator did not beat the best fixed setting live.
  - The simulator was accurate; what failed was the learner's search. A simple rule found headroom the learner did not.
  - Most of the large-range advantage under the cap comes from the arrival-time rule.

9D. CAP-TRAINED DQN V2: BROADER TRAINING, TESTED ON HELD-OUT SCENARIOS

A second predeclared study (`testdata/rl_cap_train_v2/protocol.md`) kept the learner, actions and state of §S9b. It changed the training distribution and the calibration:

- cap C 4–24 MiB/s;
- one-way delay 10–120 ms;
- early and late rate dips;
- per-request timing calibrated at four delays.

An independent review revised the protocol before it was frozen. Four new held-out scenarios lie **inside** the broadened ranges, as new combinations:

- H_C12: C = 12 MiB/s from 6 s;
- H_RTT110: 110 ms each way, cap from 12 s;
- H_DIP: cap from 6 s plus a dip from 30 to 40 s;
- H_C20: C = 20 MiB/s from 4 s, 20 ms each way.

Each had 10–17 % simulator headroom between the best fixed N and a 16 → n oracle.

- **Calibration and blind check (Confirmed for the simulator).**
  - 24 calibration runs showed client overhead rising with delay.
  - Pooled rate classes mispredicted one in-sample cell. Four classes were adopted before the blind check (amendment A1).
  - All four blind predictions at 80 and 120 ms came within ±8 % (+1.4 %, −3.3 %, +5.6 %, +5.6 %), including one that ran through the new custom-schedule path.
- **Training.**

  - 5 seeds of about 320,000 steps each. Seed 1 was selected on validation alone.
  - Disclosed before any live run (amendment A3): the other four seeds were mostly at or above the best fixed time on the held-out scenarios.
- **In distribution (Confirmed, 8 blocks, 64 runs, all valid).**
  - CAP8: v2 took 0.845 [0.840, 0.850] of the time of fixed 16 × 1 MiB, and 0.929 [0.917, 0.944] of v1.
  - Without a cap it was slower than fixed 16 × 1 MiB (1.047 [1.043, 1.052]).
  - The C=8-tuned drop rule was faster than v2 in both scenarios (CAP8: v2 ÷ drop 1.048).
- **Held out (Confirmed, 8 blocks, 160 runs, all valid).** v2 against the better fixed arm:
  - H_C12: 0.882 [0.877, 0.886], faster;
  - H_RTT110: 0.819 [0.795, 0.848], faster;
  - H_DIP: 0.952 [0.945, 0.958], faster;
  - H_C20: 1.007 [1.000, 1.014], slower by the bootstrap. The t-interval [0.998, 1.016] does not separate it from 1, and by the protocol the less favourable verdict counts.
  - v1 on the same four was faster in 1 and slower in 2. v2 was faster than v1 in all 6 live scenarios.
  - In H_RTT110, v2 was 10 % faster live than the simulator predicted (68.6 s against 76.4 s), while the fixed arms were 4 % slower. That advantage is larger than the simulator supports, and is not explained here.
- **Against the C=8-tuned drop rule:** v2 was faster in 3 scenarios (H_C12 0.968, H_RTT110 0.746, H_C20 0.743), slower in 2 (CAP8, HEALTHY) and not distinguishable in 1 (H_DIP). The rule's simulated time was already worse than fixed 16 × 1 MiB in H_RTT110 and H_C20, so two of those three wins reflect the rule's tuning to C = 8 MiB/s more than learning.
- **Reading, as frozen:**
  - "v2 generalized partially: faster in 3, slower in 1 of 4 held-out scenarios, all inside the v2 training ranges (v1 on the same scenarios: faster in 1, slower in 2)".
  - "Learned control adds something over a C=8-tuned drop rule: faster in 3, slower in 2 of 6 scenarios", with the caveat above.
- **Scope.** The comparison is with the best fixed setting at the same 1 MiB range size, under server-side changes, on one server model. It is not with 16 × 32 MiB, aria2 or real servers. This is one selected seed from a procedure that is not reliably good across seeds. It costs about 5 % when nothing changes.

9E. SYNTHESIS: ADAPTIVE ARMS AGAINST ARIA2 UNDER CHANGING CONDITIONS

This section answers the question of §S1 directly, from results reported above. Each row is a scenario in which conditions change during the download. Ratios are time ÷ aria2's time; below 1, ours was faster.

- **When the server rate rose,** every adaptive arm was faster than aria2, and the fixed configuration with the same small ranges was faster still. The gain over aria2 there comes from range granularity and request timing (§S4), not from adapting.
- **When the server degraded,** aria2 was faster than every adaptive arm. Its large pieces start before the cap and keep the pre-cap rate (§S7a); on the corrected engine our best fixed configuration came within 1.6 % of it.
- **Comparability.** The rows come from different studies, files and aria2 configurations, as noted in the sections cited; the trajectory and W1–W6 rows used the original engine. The table is a guide to direction, not a pooled estimate.
- **Not tested.** No study changed loss or delay during a download on a real path with adaptive arms running. In the correlated-loss screen the best configuration did not move with loss (N* = 16 and, without the 30 s deadline, S* = 16 MiB in all variants), so by the headroom-first method there was nothing for a controller to track; a live test is future work.

10. DISCUSSION, LIMITATIONS, AND THREATS TO VALIDITY

- **Constructed environment.** The positive adaptive results hold under a server model we designed and selected (72-candidate search). They show conditions under which adaptation can help, not that real servers behave this way.
- **Request-start rate semantics** drive both the D2 mechanism and the advantage of large ranges under the cap (§S7a). The §S9c control confirmed this for the cap: on a server that re-checks rates during each response, fixed 16 × 32 MiB slowed from 50.9 s to 67.4 s.
- **Learned-policy generalization is narrow.** The v2 result (§S9d) holds for held-out scenarios inside the training ranges, for one seed selected on validation from a procedure that was not reliably good across seeds, and against the best fixed setting at the same range size only.
- **Engine before policy.** On our server models, the request deadline and straggler splitting mattered more than any adaptive policy (§S7a). Earlier adaptive results measured against the original engine (§S6, §S7) should be read with that comparison in mind. Removing the deadline means a fully stalled connection would wait for the run limit; a stall-based timeout is future work.
- **Single machine, single aria2 version (1.37.0), loopback + netem** for every controlled result. Real paths appear only in the descriptive live pilot of §S8a: one client, one primary access network (plus one accidental slot on a second), public mirrors, and fixed cells only; the later no-harm check ran the v2 agent and the drop rule on three of those mirrors (5 blocks; both were slower than the best fixed setting there). That pilot is not a validation across networks. The only stock-server check is the NGINX per-response rate-limit screen of §S4.2 (n=5 per configuration), and it exercised neither a shared cap

What changes mid-download	Our adaptive arms ÷ aria2	Our best fixed configuration ÷ aria2
Server rate dips, then rises (D2)	rule-based joint (Track 1) 0.876	16 × 4 MiB 0.376 (§S4)
Rate rises at 1 s, no cap	frozen DQN v6 0.359; online tuner 0.432; joint agent 0.388	16 × 1 MiB 0.344 (§S7a)
Cap switches on at 8 s	causal 1.096 (original engine); cap-trained v1 1.310; frozen v6 1.367; joint agent 1.422; online tuner 1.781	16 × 32 MiB 1.016 (§S7a)
Bandwidth collapse, latency spike, burst loss, throttle (trajectories A2–E2)	DQN v6 1.456 (one of five trajectories not slower)	not run
Six capped workloads (W1–W6)	workload selector 1.660	not run

nor a connection limit, so the adaptive results of §S6-§S7 have no stock-server counterpart.

- **Replayed traces vary capacity only.** The cellular replays added no loss or propagation delay and no server behaviour (§S8). Loss and delay were studied at fixed levels with netem, and changing them during a download was covered only by the early trajectories on our own server; no live test with loss or delay changing mid-download was run.
- **Fairness.** Opening 16 connections takes a larger share of a shared link from other flows and loads public mirrors more; we measured speed, not fairness.
- **Statistics:** block-level intervals reflect emulator noise; workload-level intervals are primary for cross-workload claims; n = 5 sweep pairs are screening only.
- **Multiplicity and amendments:** the W1–W6 family was selected from 72 analytic candidates and its manifest was amended twice before the live runs; the protocol was predeclared in a logged protocol, but not independently timestamped (§S6.1 and the note above).
- **Failed gates are reported as failed** (§S6 S-margin; §S8 trace gate).
- **A bounded model-based controller attempt was stopped at its own predeclared simulation gate** and produced no live evaluation (`reports/model_based_attempt.md`); the adaptive results of §S6–§S7 therefore rest on frozen selection rules and a threshold controller, not on a controller that identifies the server model online.

11. CONCLUSION

A fixed 16×4 MiB configuration beats aria2 by a large margin on one phase-locked-pacing workload; controls attribute much of that margin to granularity, request-start timing and pacing, it shrinks when granularity is matched, and a five-pair screen across six pacing schedules shows the direction holds in all but a constant slow-rate schedule where the two are at parity. On recorded cellular traces replayed in a controlled setup there is no exploitable concurrency headroom. Adaptive control helps within a download only under a server model in which rate depends on concurrency, and only against small-range fixed configurations on the original engine; once a request deadline and straggler splitting were corrected, the best fixed configuration matched or came within 1.6 % of aria2 and beat the adaptive controllers and an online tuner. A DQN trained on a blind-checked simulator beat the best fixed setting at its own range size where it was trained, and, with broader training, on 3 of 4 held-out scenarios inside its training ranges, while a simple drop rule remained competitive and a joint range-size agent did not find the larger gains its simulator offered; a cross-workload selector matches the per-workload best on average but a consistent advantage over a single family-wide configuration is not established; learned policies were slower than aria2. Against aria2 under changing conditions, adaptive arms were faster only when the server rate rose, where a fixed configuration of the same engine was faster still, and slower whenever the server degraded. These results delimit when adaptation is worth its complexity; they do not establish general superiority of adaptive downloading. Future work: action spaces with delayed, changing costs (multi-mirror scheduling, restarting slow requests, straggler hedging), a live test with loss and delay changing during the download, and measurements across many clients and networks, each behind a headroom gate and against both the best static configuration and a strong myopic heuristic.

DECLARATIONS

Authors. Sourajyoti Basak, Shwetank Shrijan and Ayush Konar, Vellore Institute of Technology, Vellore, India (course project, Computer Networks Lab, BCSE308P).

Author contributions (as recorded in the Review 2 report). S. Basak: downloader architecture and core implementation; adaptive concurrency and range management; test server and Linux network experiments; system integration and validation. S. Shrijan: downloader development and testing of assigned components; monitoring and experiment automation; baseline and heuristic evaluation. A. Konar: LinUCB; RL state, action and reward design; Python/PyTorch models; integration of the learned policy with the Go downloader; ML/RL evaluation.

Author order, corresponding author, ORCID, funding, conflicts of interest: TO BE CONFIRMED BY THE AUTHORS.

Use of AI tools (draft; authors must confirm wording and scope). An AI coding assistant (Anthropic Claude, run through the Claude Code CLI) was used during the project to write and modify source code and analysis scripts, to run experiments and tests, to audit results against raw data, to correct manuscript claims, and to draft this text. It

did not generate experimental data other than by executing the specified programs. All numerical claims are recomputed from raw run files by `tools/claims_check.py`. Human authors are responsible for the content and must verify this disclosure against the actual workflow and the publisher's policy.

Data and code availability. Code: MIT license (copyright holder to be confirmed). Raw runs, manifests and analysis outputs are under `testdata/`; because of size they are distributed as a separate archive with SHA-256 checksums (`docs/ARCHIVE_PLAN.md`). **DOI: not yet issued.** Third-party trace files keep their own terms (`DATA_LICENSE.md`).

Reproducibility. `docs/REPRODUCING.md`; `python3 tools/claims_check.py` recomputes every quoted number; protocols and hashes for the pacing sweep and the NGINX check are committed before their runs.

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